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Department of Computer Science and Engineering
Program: Bachelor of Science in Computer Science and Engineering

Project Proposal

Course No: CSE-1200

Course Title: Software Development – I

Project Title: **Afterburn – No Escape**

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Submitted to

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Introduction

Game Description:

Afterburn – No Escape is a fast-paced, top-down arcade car chase game inspired by the classic arcade game: Highway Pursuit. The player takes control of a getaway vehicle speeding through endless highways while being relentlessly chased by police forces. The game focuses on high-speed reflexes, strategic lane switching, aggressive takedowns, and survival under increasing pressure.

As the chase escalates, enemy vehicles become more dangerous, the road grows more chaotic, and the player must rely on skill and quick decision-making to stay alive.

Objective:

The objective of this project is to develop an intense and addictive arcade pursuit game that challenges players' reaction time, spatial awareness, and risk management while delivering continuous adrenaline-driven gameplay.

Target Audience:

The game is designed for casual and arcade-style gamers aged **12 and above** who enjoy fast-paced action games, endless runners, and car chase mechanics.

Game Concept and Features

Gameplay Description:

- The player controls a car moving along a multi lane highway.
- Enemy vehicles (police cars, armored vehicles, roadblocks etc.) attempt to stop the player.
- The player has to dodge civilian traffic and ram enemies and collect powerups.
- The game continues endlessly until the player's health runs out.

Core Features

- **Highway Pursuit Mechanic:**
 - Constant forward motion with lane-based movement.
 - Increasing speed as score increases.

- **Enemy System:**
 - Police cars that chase and ram the player.
 - Special enemy units appear after score milestones.
 - Enemy difficulty scales with progression.
- **Score System:**
 - Score increases with distance traveled and enemies destroyed.
 - Bonus points are awarded for near misses.
- **Star / Heat System:**
 - A star counter represents the intensity of the police chase.
 - Higher stars introduce tougher enemies and obstacles.
- **Boss Encounters:**
 - After reaching specific score intervals, boss vehicles appear.
 - Boss fights loop with increased difficulty each time.

Innovative Elements

- **Dynamic Highway Environment:**
 - Road layouts change over time with traffic density, obstacles and scenery variations.
- **Story Illustration:**
 - Short story panels appear during the beginning and during milestone to give context to the chase.
- **Risk Reward System:**
 - Players are encouraged to drive aggressively for higher scores but at greater danger.
- **Leaderboard System:**
 - Player can enter their name and compete for the highest score in a classic arcade fashion.

Timeline

Week	Task
Week 1	Game concept finalization, reference study and asset planning.
Week 2	Implement vehicle movement and dynamic background.
Week 3	Add traffic and enemies and implement collision detection
Week 4	Implement logic for star / heat system.
Week 5	Implement specific enemy behavior after reaching milestone.
Week 6	Define scoring and powerups system.
Week 7	Implement boss battle loop and polish level design.
Week 8	Final debugging, optimization and documentation preparation.
Week 9	

Inspiration

This game is inspired by a classic arcade pursuit game known as Highway Pursuit (<https://highway-pursuit.en.uptodown.com/windows>).

Some photos of reference game:



Conclusion

This proposal outlines the development plan for **Afterburn – No Escape**, a high-speed arcade chase game designed to deliver nonstop action and escalating tension. With dynamic enemy systems, boss encounters, and competitive leaderboards, the game aims to provide an engaging and replayable experience that keeps players pushing their limits on the highway.